LLM Judge and Biomedical Expert Agreement in Alzheimer's Disease Literature RAG: An Exploratory Query-Reformulation Study

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Abstract

Automated large language model (LLM) judges are widely used to evaluate retrieval-augmented generation (RAG), but their scores may not align with domain-expert ratings. We report a post-hoc exploratory comparison of two scoring protocols in an Alzheimer's disease literature-RAG pipeline. Eight configurations combined BM25 or DPR retrieval with keyword prompting and hypothetical document embeddings (HyDE). Five biomedical researchers who had generated the query pool rated responses for 50 queries on factuality, diversity, clarity, and insightfulness; GPT-4o provided secondary ratings. The protocols were not equivalent: experts reviewed retrieved abstracts and conditions side by side, whereas GPT-4o received each query–response pair individually without abstracts. Condition-mean expert scores ranged from 3.17 to 4.62 and GPT-4o condition means from 3.14 to 4.34. Pooled query–condition Pearson correlations were 0.28–0.54 and Spearman correlations were 0.27–0.52. Single-rater intraclass correlations were 0.09–0.24. The protocols showed modest query–condition-level agreement and systematic scale differences. Because evaluator type was confounded with evidence access and presentation format, the differences cannot be attributed solely to human-versus-LLM judgement. Automated ratings should complement rather than replace expert evaluation in this setting.

1 Background

Retrieval-augmented generation (RAG) conditions language-model responses on external documents and offers a practical way to support biomedical literature question answering [1–3]. Domain terminology and compositional information needs make retrieval challenging. Query rewriting and hypothetical document generation are established ways to modify retrieval inputs [4, 5]; biomedical systems also explore generative query expansion and iterative follow-up questions [6, 7]. These techniques motivate our archived Alzheimer's disease literature-RAG experiment, but response quality must be evaluated separately from retrieval effectiveness.

LLM judges make response evaluation scalable, yet their agreement with people depends on the task and protocol. Prior work on MT-Bench and Chatbot Arena examined agreement with human preferences and biases in automated judging [8]; subsequent work surveys reliability, consistency, and bias across evaluation settings [9]. Agreement observed in general conversational benchmarks cannot establish agreement with biomedical researchers assessing specialised literature responses. Moreover, a correlation between two sets of scores measures association, not equality of scores or factual correctness.

Shared query pool: 201 researcher-authored queries; 50 with expert ratings.

Eight configurations: BM25 or DPR, each with naive retrieval, HyDE, keyword HyDE, or MeSH-guided keyword HyDE.

Evidence and generation: two retrieval steps per stream; two document slots without keywords, usually four with keywords; GPT-4o-mini response generation.

Expert protocol: five researchers; query, response, and abstracts; conditions shown side by side.

GPT-4o protocol: query and response; no abstracts; one response at a time.

Analysis: 400 query–condition pairs; four scoring dimensions; query-cluster bootstrap and expert inter-rater reliability.

Figure 1: Study design and the two realised scoring protocols. Retrieval uses BM25 or DPR separately. MeSH is prompt guidance, and the two-step procedure does not implement multi-hop reasoning. Evidence access and presentation format differ between evaluation protocols.

Despite these advances, controlled comparative studies examining multiple reformulation strategies within a specific biomedical setting remain limited. This study conducts a post-hoc exploratory analysis of an Alzheimer's disease literature-RAG experiment. The recovered automated-evaluation archive contains keyword generation (key-concept-based and MeSH-guided prompting), query rewriting, and pseudo-document generation (HyDE) conditions. The comparison is restricted to the eight conditions for which both expert and automated ratings are available. Because the expert and GPT-4o protocols differed in evidence access and presentation format, our analysis estimates agreement between two realised scoring protocols rather than isolating an effect of evaluator type. Our contributions are: (i) an empirical comparison of expert and automated ratings on real researcher queries; (ii) an analysis of agreement, directional scoring differences, and expert inter-rater reliability; and (iii) a transparent audit of protocol, design, and implementation limitations that constrain method-level conclusions.

2 Methods

We retrospectively analysed archived responses and ratings from a query-reformulation experiment. The primary estimands are associations and mean score differences between expert-panel averages and GPT-4o ratings on the same query–condition pairs. Configuration scores provide context for this comparison; the design does not isolate causal effects of retrieval or reformulation choices.

For each of BM25 and DPR, the shared archive includes four configurations: the original query without keywords (naive), HyDE without keywords, key-concept keyword prompting with HyDE, and MeSH-guided keyword prompting with HyDE. Rewriting conditions are described for archive provenance but excluded from the paired analysis because expert ratings are unavailable.

2.1 Keyword Generation

To more precisely capture the information needs inherent in a query, we introduce an initial step that automatically generates core keywords using an LLM. However, keywords generated by an LLM may be overly generic or misaligned with expert-level biomedical concepts. We therefore design two complementary keyword generation strategies with distinct objectives. This process can be formalized as follows in (1):

$$\mathrm{LLM}(q) \rightarrow k \tag{1}$$

where q denotes the input query and k represents the set of keywords generated by the LLM.

- **Based on Key Concepts:**
  The first strategy selects the essential concepts required to understand and resolve the query by considering its semantic structure and context. This approach leverages the LLM to generate keywords that directly reflect the core information needs of the user. Overly generic or ambiguous terms such as relationship or mechanism are excluded, while biologically meaningful concepts with direct domain relevance are prioritized.

- **MeSH-Guided Prompting:**
  The second strategy provides the LLM with prompt-level guidance to prefer biomedical terminology consistent with MeSH (Medical Subject Headings) vocabulary, prioritising terms related to genes, diseases, and biochemical pathways. Importantly, this condition did *not* perform explicit MeSH ontology matching, MeSH-term retrieval from a controlled vocabulary list, or terminology expansion. The prompt instructed the LLM to prefer MeSH-style terminology; we therefore term this approach *MeSH-guided prompting* rather than ontology-based query expansion, and its results should be interpreted accordingly.

These keyword generation strategies identify the semantic core of the query and provide a foundation for subsequent query transformation.

The prompt requested up to two keywords per query. Inspection of the recovered 201-query archive showed that 199 queries produced two keywords, while two queries produced five; the implementation retained all returned keywords rather than truncating them. Both exception queries were included in the 50-query human-evaluation subset. We therefore describe two keywords as the intended output, not an invariant of the realised experiment.

2.2 Query Transformation

When the original query is used directly for retrieval, deficiencies in its expression or semantic specificity often lead to degraded retrieval performance, which subsequently negatively affects the response generation stage in RAG systems[10]. To mitigate this issue, we adopt two query transformation strategies, as formalized in (2):

$$\mathrm{LLM}(q, k) \rightarrow q^* \tag{2}$$

where q^* denotes the transformed query.

- **Keyword-Focused Query Rewriting:**
  Query rewriting reformulates the original query into a form better suited for retrieval. Following Ma et al. [4], we employ an LLM to perform rewriting. Unlike prior work, our approach incorporates not only the original query but also the previously generated keywords, anchoring the rewritten query around them to provide clearer semantic guidance. The rewriting prompt instructs the model to produce a keyword-focused reformulation; it does *not* include an explicit ambiguity-detection or ambiguity-resolution component. This method should therefore be interpreted as keyword-focused query rewriting rather than ambiguity-aware rewriting.

- **Pseudo-Document Generation:**
  This strategy generates a pseudo-document from the query, mapping the query representation closer to the document embedding space and facilitating more effective semantic alignment. While Gao et al. [5] generate pseudo-documents solely from the original query, our method additionally conditions the generation on the extracted keywords. This conditions pseudo-documents on the selected concept; whether it improves retrieval alignment was not measured directly.

Through this sequence of steps, we derive transformed query representations and perform retrieval based on them.

2.3 Two-Step Context-Augmented Retrieval

Even with retrieval-optimised query representations, retrieving only a single document may fail to satisfy complex, multi-concept queries. To provide broader evidence coverage, we employ a two-step retrieval procedure based on the transformed query q^*, formalised in Equation (3). Here, d_1 and d_2 denote documents retrieved at each step.

$$\begin{aligned} \text{Retriever}(q^*) &\rightarrow d_1 \\ \text{Retriever}(\{q^*, d_1\}) &\rightarrow d_2 \end{aligned} \tag{3}$$

For each transformed-query stream, we retrieve the top-ranked document d_1 for q^* ($k = 1$). We then concatenate q^* and d_1 to form an expanded context, which is used to retrieve a second document d_2. In keyword-conditioned implementations, this procedure is run independently for every returned keyword. No-keyword conditions therefore supplied two document slots to the generator, whereas keyword conditions usually supplied four slots. The two five-keyword exceptions supplied ten slots. Exact duplicates could remain across keyword streams.

Within a stream, the intended rule was to select the next-ranked result when the second-step top result duplicated d_1. A code audit identified an exception in the recovered BM25 no-keyword implementation: the second-ranked candidate was selected in both branches of the conditional, including when the top-ranked candidate was not a duplicate. We analyse the archived outputs as generated and do not retroactively correct this implementation artefact. No sub-question decomposition, evidence chaining, or multi-hop reasoning is performed.

2.4 Generation

Following the retrieval stage, we generate the final response using an LLM. The generation process can be formalized as follows:

$$\text{LLM}(q, K, D) \rightarrow a \tag{4}$$

where q denotes the original query, K represents the set of generated keywords, D denotes the collection of documents obtained through two-step context-augmented retrieval, and a denotes the generated response.

Unlike naïve RAG approaches that provide only the query and retrieved documents as input [1, 10, 11], the keyword-conditioned implementation jointly incorporates q, K, and all retrieved documents D into the LLM. Keyword conditions therefore differ from no-keyword conditions in both prompt content and evidence budget. Method-level score differences cannot be attributed solely to query reformulation.

2.5 Experimental Setup

Query Dataset

We evaluated a subset of 50 queries drawn from a recovered pool of 201 unique queries generated by the five biomedical researchers who later served as evaluators. Earlier project notes described the pool as 200 queries; the value 201 reported here follows the archived files (IDs 0–200). Queries include questions posed to large language models during ongoing research as well as anticipated future questions, spanning disease pathogenesis, biochemical pathways, cellular mechanisms, transcriptional regulation, diagnostic markers, experimental model comparisons, and therapeutic strategies. The procedure used to select the 50-query human-evaluation subset is not fully documented, so it should not be assumed to be a random or representative sample.

Knowledge Base

To construct the external document corpus, we collected PubMed article abstracts spanning 2015–2024 using the Biopython Entrez API. We focused on recent publications and used abstracts rather than full texts, consistent with prior findings that shorter documents reduce input-context costs and can yield higher retrieval recall [12]. Literature retrieval used the search term "Alzheimer's disease". The abstracts were processed into text chunks prior to embedding. The corpus comprised approximately 18,000 PubMed abstracts after collection; exact deduplication procedures and chunking parameters are not fully recovered from available records and represent a reproducibility limitation.

Implementation Details

For retrieval, we employed BM25 [13] as a sparse retriever using default parameters (without specific tuning for k_1 or b) and DPR [14] as a dense retriever using the `bert-base-encoder.cp` checkpoint. For response generation, we used GPT-4o-mini as the generation model. For keyword generation and query transformation, GPT-4o-mini was used with temperature = 0 (with all other parameters left at default) and a fixed output format specified in the prompt. No post-hoc candidate ranking, semantic filtering, or hard truncation of excess keywords was applied. Duplicate handling was performed within each two-step stream; duplicates could remain across keyword streams. Retrieval scores and PubMed identifiers were not retained in the evaluation archive.

2.6 Evaluation Metrics

We evaluated the quality of generated responses using the following four dimensions, assessed on a five-point Likert scale (1 = very poor, 5 = excellent) by both human experts and an automated LLM judge.

- **Factuality**: Whether the response provides factually accurate information without introducing false statements with respect to the given query.
- **Diversity**: Whether the response reflects a broad range of information, perspectives, approaches, or aspects in addressing the given query.
- **Clarity**: Whether the response is clearly structured, logically organized, and easy to understand.
- **Insightfulness**: Whether the response goes beyond surface-level information and offers meaningful insights or perspectives that help interpret or resolve the query.

2.7 Human Evaluation Protocol

Five biomedical researchers associated with ongoing Alzheimer's disease research independently evaluated the 50-query subset. These same five researchers had generated the query pool. Each evaluator received a structured Excel file containing all eight reformulation conditions for each query, presented side by side to facilitate within-query comparison. For each condition, evaluators were shown the original query, the generated response, and the archived retrieved abstracts for that condition: two slots for no-keyword conditions, usually four for keyword conditions, and ten for the two five-keyword exceptions. Evaluators assigned integer scores (1–5) on each of the four dimensions using the rubric described above. No evaluator was given information about which reformulation strategy produced a given response. Condition order randomisation and formal rater training are not documented.

The evaluation was conducted after the automated GPT-4o evaluation had been completed; it was not pre-registered as the primary analysis. Accordingly, the recovered human-evaluation

records were analysed as a post-hoc exploratory evaluation. The human results are interpreted as exploratory evidence rather than confirmatory evidence. There were 2,000 query–condition–rater records and 8,000 possible scalar scores. A missing query-text cell in one evaluator file was linked by its query identifier after verifying that all eight response and context texts matched the other files; its available ratings were retained. Missing scores were excluded without imputation; missingness and complete-case sample sizes are reported in Table 2. Evaluator-level data are released only under stable pseudonymous identifiers.

2.8 Automated Evaluation

Automated evaluation was conducted using GPT-4o [15] as an LLM judge, assigning scores on the same five-point Likert scale under each metric. For each rating, GPT-4o received the original query and generated response but was not shown the retrieved abstracts. Responses were scored individually rather than in the side-by-side condition display used for expert evaluation. Thus, evaluator type was confounded with evidence access and presentation format. We note that GPT-4o is from the same model family as the generator (GPT-4o-mini), which may introduce a same-family preference bias that inflates absolute scores or favours stylistically similar outputs [9]. The judge prompt, rubric, and exact scoring instructions are shown in the Appendix. All automated results should be interpreted as descriptive comparisons between the two realised protocols, not as an isolated test of human-versus-LLM judgement.

2.9 Agreement Analysis

To quantify agreement between the expert and automated scoring protocols, we computed Pearson and Spearman correlations between the available-rater mean and GPT score over the 400 query–condition pairs for each metric. Each pair receives equal weight. Human condition means in Table 1 instead pool available individual ratings, so their weighting differs slightly when scores are missing. Correlation is an association measure; mean human-minus-GPT differences describe scale offsets. These correlations should not be interpreted as isolating evaluator identity because the protocols differed in evidence access and presentation format. Condition-level ranking agreement was assessed via Spearman rank correlation across the eight conditions using full-precision condition means from Tables 1 and 3, with average ranks for ties. Inter-rater reliability among the five experts was assessed using the intraclass correlation coefficient (ICC(A,1): two-way random effects, absolute agreement, single rater) and Krippendorff's α (interval scale). Both reliability estimators used complete five-rater items, separately for each metric. Treating the ordinal scores as interval values is an analytic assumption; Spearman correlation provides a rank-based companion measure. Repeated queries across conditions and a fixed expert panel limit generalisation of these reliability point estimates. To account for repeated observation of each query across eight conditions, percentile 95% confidence intervals were calculated using a query-cluster bootstrap with 2,000 replicates (seed: 20260807); every resampled query retained all eight conditions. These intervals reflect query-sampling uncertainty after averaging experts, not uncertainty from sampling a different panel of evaluators. Analyses used the `pingouin`, `krippendorff`, and `scipy` packages. Because the evaluation was post-hoc and exploratory, no multiplicity-adjusted confirmatory tests are presented.

3 Results

3.1 Human Evaluation Results

Table 1 reports descriptive average expert scores across the eight shared conditions. These are exploratory aggregate results; given low inter-rater agreement, they should not be interpreted as stable ground-truth estimates. Under BM25, keyword-augmented HyDE conditions tended to

Table 1: Descriptive average expert scores (5 evaluators, 1–5 Likert scale) for the eight shared conditions. Values pool all available evaluator scores within each condition (up to 250 per metric); missing scores are excluded. These are exploratory observations; low inter-rater agreement (ICC(A,1): 0.09–0.24) limits interpretability.

Condition	Factuality	Diversity	Clarity	Insightfulness
Retriever: BM25				
HyDE (no keyword)	4.50	3.48	3.72	3.44
Naive (no keyword)	4.30	3.27	3.40	3.17
HyDE + Keyword	4.62	3.64	4.04	3.61
MeSH-guided + HyDE	4.53	3.67	4.02	3.53
Retriever: DPR				
HyDE (no keyword)	4.37	3.51	3.51	3.27
Naive (no keyword)	4.32	3.40	3.39	3.17
HyDE + Keyword	4.39	3.44	3.67	3.41
MeSH-guided + HyDE	4.35	3.64	3.66	3.42

Table 2: Missing scalar scores and expert inter-rater reliability. Each metric has 2,000 possible scores; reliability uses items with all five ratings. ICC(A,1) is single-rater absolute agreement; ICC(A,5) is agreement for the panel mean. Alpha uses interval distances. These are descriptive point estimates.

Metric	Missing	Items	ICC(A,1)	ICC(A,5)	α
Factuality	15	387	0.09	0.34	0.02
Diversity	15	387	0.24	0.61	0.21
Clarity	16	386	0.15	0.47	0.11
Insightfulness	15	387	0.11	0.38	0.07

receive the highest expert scores on factuality ($\bar{x} = 4.62$) and diversity ($\bar{x} = 3.67$). Under DPR, differences across conditions were smaller in magnitude. Expert scores were notably higher on factuality (overall $\bar{x} = 4.42$) than on insightfulness ($\bar{x} = 3.38$). This difference in mean score should not be interpreted as greater consistency: factuality had the lowest single-rater ICC.

Inter-rater agreement was low for individual raters across all four metrics: ICC(A,1) ranged from 0.09 (factuality) to 0.24 (diversity), and Krippendorff's α ranged from 0.02 (factuality) to 0.21 (diversity). These values indicate that individual expert ratings varied substantially even for the same response, which substantially limits the interpretability of condition-level differences. Descriptive differences across conditions should be understood as reflecting this uncertainty.

3.2 Automated GPT Evaluation Results

Table 3 reports GPT-4o judge scores for the same 50-query subset and eight conditions. GPT scores on clarity were consistently higher than expert scores (GPT $\bar{x} = 4.02$–4.34 vs. expert $\bar{x} = 3.39$–4.04), while GPT scores on factuality were consistently lower than expert scores (GPT $\bar{x} = 3.64$–4.00 vs. expert $\bar{x} = 4.30$–4.62). Under BM25, MeSH-guided HyDE achieved the highest GPT score on diversity (3.64); under DPR, keyword HyDE showed the highest factuality (3.74) and clarity (4.24) scores.

We note that GPT-4o (judge) and GPT-4o-mini (generator) are from the same model family, which may introduce a same-family preference bias, particularly for clarity and style-related dimensions [9]. Absolute GPT scores should therefore be interpreted with caution. These results are presented as a secondary, descriptive analysis.

Table 3: Descriptive average GPT-4o judge scores (1–5 Likert scale) for the eight shared conditions on the 50-query subset. These results are exploratory and may be subject to same-family model bias.

Condition	Factuality	Diversity	Clarity	Insightfulness
Retriever: BM25				
HyDE (no keyword)	3.86	3.42	4.22	4.00
Naive (no keyword)	3.78	3.14	4.02	3.76
HyDE + Keyword	3.98	3.56	4.32	4.00
MeSH-guided + HyDE	4.00	3.64	4.34	3.98
Retriever: DPR				
HyDE (no keyword)	3.64	3.14	4.14	3.76
Naive (no keyword)	3.66	3.14	4.10	3.84
HyDE + Keyword	3.74	3.50	4.24	3.96
MeSH-guided + HyDE	3.70	3.46	4.18	3.88

Table 4: Agreement and directional score differences between expert averages and GPT-4o. Confidence intervals are percentile intervals from 2,000 query-cluster bootstrap replicates; all eight conditions were retained within a resampled query. H–G denotes the mean human-minus-GPT score.

Metric	Pearson r [95% CI]	Spearman ρ [95% CI]	H–G mean [95% CI]
Factuality	0.28 [0.13, 0.43]	0.28 [0.13, 0.42]	0.63 [0.44, 0.81]
Diversity	0.54 [0.43, 0.64]	0.52 [0.40, 0.63]	0.13 [0.02, 0.24]
Clarity	0.30 [0.17, 0.42]	0.27 [0.14, 0.40]	-0.52 [-0.63, -0.40]
Insightfulness	0.39 [0.29, 0.48]	0.37 [0.28, 0.45]	-0.52 [-0.61, -0.44]

3.3 Agreement and Disagreement Between Human and GPT Evaluations

Table 4 summarises the query–condition-level correlation between expert average scores and GPT scores across the eight shared conditions (50 query clusters; 400 query–condition pairs). Pearson r values ranged from 0.28 (factuality; query-cluster 95% CI: 0.13–0.43) to 0.54 (diversity; 0.43–0.64), and Spearman ρ values ranged from 0.27 to 0.52. These values indicate modest agreement across query–condition pairs.

Condition-level ranking agreement was higher: Spearman ρ across the eight conditions was 0.93 for clarity, 0.86 for insightfulness, 0.71 for diversity, and 0.67 for factuality. Thus, while GPT and human experts broadly agreed on the *relative ordering* of conditions by clarity and insightfulness, their ordering of conditions by factuality was less consistent.

Directional differences were also observed. Averaged across the shared conditions, the human-minus-GPT difference was +0.63 points for factuality (query-cluster 95% CI: 0.44–0.81), +0.13 for diversity (0.02–0.24), −0.52 for clarity (−0.63 to −0.40), and −0.52 for insightfulness (−0.61 to −0.44). The sign was consistent across the eight conditions for factuality, clarity, and insightfulness. These differences show that the two evaluation sources used the scale differently; they do not establish which source was more accurate.

4 Discussion

How Closely Did the Two Evaluation Protocols Agree?

The central finding of this study is that query–condition-level agreement between the GPT-4o and biomedical-expert scoring protocols was modest (Pearson r: 0.28–0.54; Spearman ρ: 0.27–0.52 across the four metrics). The ordering of conditions was more consistent for clarity and insightfulness ($\rho = 0.93$ and 0.86, respectively) than for factuality ($\rho = 0.67$). However, the

experts reviewed retrieved abstracts and compared all conditions side by side, whereas GPT-4o received each query–response pair individually and without the retrieved abstracts. Evaluator identity was therefore confounded with evidence access and presentation format. The observed rank agreement is descriptive and does not establish how humans and an LLM judge would compare under a matched protocol.

GPT-4o assigned higher clarity and insightfulness scores and lower factuality scores than the expert-panel average. These differences could reflect scale use, evidence access, presentation format, response fluency, or same-family model preference. No external factuality gold standard was available, and low expert inter-rater agreement (ICC(A,1): 0.09–0.24) limits claims about a collective expert preference. Accordingly, the observed differences should not be interpreted as showing that either scoring source was more correct.

Keyword Generation and Pseudo-Document Approaches

Across both human and GPT evaluations, keyword-augmented HyDE conditions tended to receive descriptively higher scores under BM25 than HyDE without keywords or naïve conditions. However, keyword conditions usually supplied four document slots, compared with two in no-keyword conditions, and two queries supplied ten. Keywords were also passed directly to the generator. The observed differences therefore combine reformulation, evidence-budget, and generation-prompt effects and cannot isolate the effect of keyword grounding.

Retriever Type and Reformulation Sensitivity

Descriptive scores differed by retriever and reformulation condition, but the design does not support a clean retriever-by-reformulation effect. In addition to the variable evidence budget, the archived BM25 no-keyword outputs were affected by the second-step selection artefact described in Methods. Retriever comparisons are therefore retained as observations about this archive rather than evidence that sparse retrieval benefits more from keyword reformulation.

Low Inter-Rater Agreement Among Experts

The low ICC values observed in this study (ICC(A,1): 0.09 for factuality, 0.24 for diversity) warrant careful interpretation. These values indicate that the five expert raters applied substantially different standards, possibly reflecting differences in disciplinary background, expectations for response length, or interpretation of the rubric dimensions. This substantially limits the strength of conclusions that can be drawn from expert ratings alone, and underscores the importance of using agreed-upon rubrics and providing rater training in future human evaluation studies. The low factuality ICC in particular is notable: despite factuality being the most concrete dimension, raters disagreed substantially on individual responses.

Limitations of the Automated Evaluation

The automated and expert evaluations did not use a matched input and presentation protocol, which prevents attribution of observed differences solely to evaluator type. In addition, GPT-4o is from the same model family as the generator (GPT-4o-mini), so same-family preference bias may inflate absolute scores or favour stylistically similar outputs [9]. GPT's higher clarity scores are compatible with, but do not prove, this hypothesis. Keywords passed to the generation prompt may also make responses appear more structured even when their substantive content is not superior.

5 Conclusion

This study reports a post-hoc exploratory comparison of two non-equivalent scoring protocols applied to query-reformulation outputs in an Alzheimer's disease literature-RAG setting. We compared eight reformulation configurations (key-concept-based and MeSH-guided keyword generation combined with HyDE, across BM25 and DPR retrievers) using ratings from five biomedical researchers and GPT-4o.

Query-level correlation between the two protocols was modest (Pearson r: 0.28–0.54), while condition-level ranking agreement was higher, particularly for clarity and insightfulness. GPT-4o assigned higher clarity and insightfulness scores and lower factuality scores than the expert-panel average. Because evaluator type was confounded with evidence access and presentation format, these differences cannot be attributed solely to human-versus-LLM judgement. Automated ratings should therefore complement rather than substitute for expert evaluation in this setting.

Within this archive, keyword-augmented HyDE conditions were associated with higher descriptive scores under BM25. Variable document counts, direct keyword input to the generator, and a BM25 baseline implementation artefact prevent a causal interpretation of that pattern. All conclusions are exploratory and limited to this single-domain, post-hoc context.

Limitations

- **Single domain.** All queries concern Alzheimer's disease research; findings may not generalise to other biomedical domains or query types.
- **Post-hoc exploratory design.** The human evaluation was not pre-registered as the primary analysis; all results are exploratory evidence and should not be interpreted as confirmatory.
- **Non-equivalent evaluation protocols.** Experts reviewed retrieved abstracts and all conditions side by side, while GPT-4o received each query–response pair individually and without abstracts. Evaluator identity is therefore confounded with evidence access and presentation format.
- **Undocumented subset selection.** The process used to select 50 of the 201 archived queries for human evaluation was not recovered, limiting claims of representativeness.
- **Low inter-rater agreement.** ICC(A,1) values of 0.09–0.24 indicate that expert ratings varied substantially across evaluators, limiting the stability of condition-level human evaluation averages.
- **MeSH-guided prompting, not ontology expansion.** The MeSH condition used prompt-level guidance; results should not be interpreted as evidence for ontology-based query expansion.
- **Two-step retrieval is not multi-hop reasoning.** The procedure retrieved one document per step within each transformed-query stream; no sub-question decomposition or evidence chaining was performed.
- **Variable evidence budget and prompt confounding.** Keyword conditions usually supplied four document slots and explicit keywords to the generator, while no-keyword conditions supplied two slots. Two queries produced five rather than two keywords and supplied ten slots. Method-level effects are therefore not isolated.
- **BM25 no-keyword implementation artefact.** The recovered implementation selected the second-ranked second-step candidate in both branches of its duplicate-handling conditional. Archived BM25 no-keyword outputs were not regenerated, so comparisons involving this baseline require caution.

- **Automated evaluation bias.** GPT-4o (judge) and GPT-4o-mini (generator) are from the same model family, which may introduce same-family preference bias, particularly for clarity and style-related dimensions.
- **No retrieval-quality metrics.** Retrieval scores, ranked candidate lists, and relevance labels were not retained in the evaluation archive. Recall@k, nDCG, and related metrics could not be reconstructed from the archived outputs alone.
- **Evaluator independence.** The five evaluators also generated the query pool. Familiarity with the questions may affect their expectations and limits generalisation to independent panels.

Future work should use a fixed document budget, correct and rerun the BM25 baseline, separate retrieval inputs from generation-prompt inputs, collect relevance labels and ranked retrieval outputs, and employ pre-registered human evaluation with rater training and independent LLM judge families.

Appendix A Prompts and Implementation Details

This appendix reproduces the prompts used in each pipeline component as screenshot images. All prompts were used with GPT-4o-mini (generation and reformulation) or GPT-4o (evaluation judge).

Recovered parameters: Temperature = 0 for keyword generation and query transformation, with all other sampling parameters left at defaults. The prompt requested up to two keywords, but the archived parser retained five for two queries. No post-hoc candidate ranking or hard truncation was applied. DPR used the `bert-base-encoder.cp` checkpoint. BM25 used default parameters (k_1, b).

A.1 Keyword Generation

Prompt: Generate General Keywords

Instruction: Generate a concise set of precise and domain-specific keywords up to `{num}` based on the given query. Focus only on the most critical and high-impact keywords that are essential for understanding or addressing the query.
Each keyword should represent the core elements of the query, such as key mechanisms, biological processes, or specific interactions. Exclude minor or overly generic terms like 'relationship' or 'mechanism' unless they are explicitly critical to the query. Prioritize terms that are unique, actionable, and directly relevant to the domain of the query.
Provide the results as a numbered list in the format: <number>. <keyword>.

Query: `{query}`

Figure A1: Prompt template for key-concept-based keyword generation (Part 1).

Prompt: Generate Mesh Keywords

Instruction: Generate a set of keywords up to `{num}` based on the given query, leveraging the MeSH (Medical Subject Headings) vocabulary. The keywords should be derived from relevant MeSH terms or closely related concepts. Reflect the core elements of the query, including biological processes, diseases, genes, proteins, and pathways. Avoid overly generic terms unless explicitly relevant to the query.
Prioritize specific MeSH terms that align directly with the query. Terms that are actionable and provide deeper insight into the topic.
Output the results as a numbered list in the format <number>. <MeSH term>.

Query: `{query}`

Figure A2: Prompt template for MeSH-guided keyword generation (Part 2).

A.2 Query Transformation

Prompt: Pseudo-Document Generation (HyDE)

Instruction: Please write down an abstract of a biology or medical paper that can solve the given query by using information or topic from abstract.

Query: `{query}`

Figure A3: Archived prompt for HyDE without keywords.

Prompt: Pseudo-Document Generation (HyDE) with Keyword

Instruction: Generate only one abstract of a biology or medical paper that can solve the given query based on the keyword. The abstract should directly address the query while focusing solely on the perspective of the keyword. Ensure that the abstract is logically structured, concise, and relevant to both the query and aspect. Provide the abstract text as plain text without titles, numbering, or additional formatting.

Query: `{query}`
Keyword: `{keyword}`

Figure A4: Archived prompt for keyword-conditioned HyDE.

Prompt: Query Rewriting

Instruction: Generate a refined and answerable question based on the given query. The revised question should eliminate ambiguity, focus on concrete and practical aspects, and ensure clarity in scope and intent. Avoid abstract or overly philosophical interpretations and instead reframe the query to facilitate a clear and actionable response.

Query: `{query}`

Figure A5: Archived prompt for query rewriting without keywords (not included in the expert comparison).

Prompt: Query Rewriting with Keyword

Instruction: Generate only one question based on the given query. The question should focus on the given keyword of the topic. Output only the question text as plain text, without any titles, numbering, or additional formatting.

Query: `{query}`
Keyword: `{keyword}`

Figure A6: Archived prompt for keyword-conditioned query rewriting (not included in the expert comparison).

A.3 Generation

Prompt: Generate without Keywords

Instruction: Using the provided abstracts, generate a response that either addresses the given query directly or provides meaningful insights that contribute to understanding or resolving the query.

Query: `{query}`
Abstracts: `{abstracts}`

Figure A7: Archived response-generation prompt without keywords.

Prompt: Generate with Keywords

Instruction: Each keyword corresponds to abstracts in one-to-one relationship. Using the provided keywords and their corresponding abstracts, generate a response that either addresses the given query directly or offers meaningful insights that contribute to understanding or resolving the query.

Query: `{query}`
Keywords: `{keywords}`
Abstracts: `{abstracts}`

Figure A8: Archived response-generation prompt with keywords.

A.4 Evaluation

Prompt: Evaluate Factuality

Instruction: Evaluate the generated response based on Factuality, which measures the extent to which the response provides accurate, truthful, and verifiable information without introducing false, misleading, or hallucinated content in relation to the given query.

Use the following 5-point scale to assign a Factuality score (1 to 5 points):

1 Point (Very Low Factuality)
- The response contains multiple factual errors, hallucinations, or clearly incorrect statements.
- It misrepresents key facts and provides false or misleading information that may significantly harm understanding or decision-making.

2 Points (Low Factuality)
- The response has some accurate content, but also includes important factual inaccuracies, unsupported claims, or misleading generalizations.
- These issues reduce the reliability of the response.

3 Points (Moderate Factuality)
- The response is mostly accurate, but may contain minor factual inaccuracies, vague statements, or claims lacking clear evidence.
- While not seriously misleading, it requires some fact-checking or clarification.

4 Points (High Factuality)
- The response is factually accurate overall, with reliable and verifiable content.
- There may be minor omissions or simplifications, but no false or misleading claims.

5 Points (Very High Factuality)
- The response is entirely accurate, well-supported by scientific evidence, and free from errors, hallucinations, or exaggeration.
- All claims are trustworthy and align with established biomedical knowledge.

Evaluation Task:
After reviewing the response, assign a Factuality score (1 to 5 points) based on the scale above. Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A9: GPT-4o evaluation judge prompt for the factuality metric.

Prompt: Evaluate Diversity

Instruction: Evaluate the generated response based on Diversity, which measures the extent to which the response integrates a broad range of relevant information, perspectives, biological mechanisms, diagnostic approaches, methodologies, or treatment strategies to effectively address the given query.

Use the following 5-point scale to assign a Diversity score (1 to 5):

1 Point (Very Low Diversity)
- The response relies heavily on a single concept, source, or perspective, with little to no exploration of other relevant aspects or alternative viewpoints.
- Critical dimensions relevant to the query, such as underlying mechanisms, diagnostic considerations, or therapeutic options, are largely absent.

2 Points (Low Diversity)
- The response mentions more than one piece of information, but coverage is narrow, repetitive, or focused on a single aspect.
- Other important aspects (e.g., additional biological pathways, diagnostic tools, or treatment approaches) are underdeveloped or missing.

3 Points (Moderate Diversity)
- The response addresses multiple relevant aspects related to the query, such as mechanisms, diagnostic approaches, or treatment strategies, but lacks depth or balance in some areas.
- There is some variety, but the range of perspectives or approaches could be expanded to provide a more comprehensive understanding.

4 Points (High Diversity)
- The response incorporates a variety of relevant biological, diagnostic, and/or therapeutic perspectives, covering most major aspects related to the query.
- The response is well-rounded, with only minor gaps or slight imbalances in coverage.

5 Points (Very High Diversity)
- The response synthesizes a comprehensive range of biological, diagnostic, and therapeutic aspects, drawing from multiple perspectives and approaches.
- All critical dimensions relevant to the query are addressed, and the information is balanced and mutually complementary.

Evaluation Task:
After reviewing the response, assign a Diversity score (1 to 5 points) based on the scale above.
Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A10: GPT-4o evaluation judge prompt for the diversity metric.

Prompt: Evaluate Clarity

Instruction: Evaluate the generated response based on Clarity, which measures the extent to which the response is expressed in a clear, well-structured, and easy-to-understand manner, helping the reader grasp the solution to the given query without confusion.

Use the following 5-point scale to assign a Clarity score (1 to 5):

1 Point (Very Low Clarity)
- The response is highly difficult to understand, disorganized, or contains excessive jargon and unclear explanations.
- The core message is obscured, making it hard to grasp how it addresses the query.

2 Points (Low Clarity)
- The response is partially understandable, but contains confusing sections, poor structure, or vague language.
- Significant effort is required to comprehend how the response relates to the query.

3 Points (Moderate Clarity)
- The response is generally understandable, but some sentences may be difficult to follow, or the structure could be improved for easier reading.
- Occasional ambiguity or complex phrasing may slightly hinder understanding.

4 Points (High Clarity)
- The response is mostly clear and easy to understand, with logical structure and minimal ambiguity.
- Minor refinements could further enhance readability, but comprehension is smooth overall.

5 Points (Very High Clarity)
- The response is exceptionally clear, concise, and well-structured, with straightforward language and smooth logical flow.
- The reader can easily understand how the response addresses the query without any confusion.

Evaluation Task:
After reviewing the response, assign a Clarity score (1 to 5 points) based on the scale above.
Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A11: GPT-4o evaluation judge prompt for the clarity metric.

Prompt: Evaluate Insightfulness

Instruction: Evaluate the generated response based on Insightfulness, which measures the extent to which the response offers meaningful, valuable, and non-obvious insights that contribute to understanding or addressing the given query.

Use the following 5-point scale to assign an Insightfulness score (1 to 5 points):

1 Point (Very Low Insightfulness)
- The response provides little to no new understanding, stating only obvious or generic information.
- It fails to offer any useful guidance in addressing the query.

2 Points (Low Insightfulness)
- The response contains some relevant information, but the insights are shallow or largely restate well-known facts.
- There is minimal value added to deepen understanding of the query.

3 Points (Moderate Insightfulness)
- The response offers a few helpful insights, but they may be somewhat basic, incomplete, or lacking originality.
- Some value is added, but the response could be more thought-provoking or detailed.

4 Points (High Insightfulness)
- The response presents several meaningful insights, going beyond surface-level information.
- It enhances understanding of the query, though a few aspects might still benefit from deeper exploration.

5 Points (Very High Insightfulness)
- The response delivers deep, valuable, and non-obvious insights, offering novel perspectives or synthesizing information in a way that greatly enhances understanding of the query.
- The response stands out as particularly insightful and well thought out.

Evaluation Task:
After reviewing the response, assign an Insightfulness score (1 to 5 points) based on the scale above. Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A12: GPT-4o evaluation judge prompt for the insightfulness metric.

Appendix B Data and Code Availability

The manuscript, analysis scripts, and text-free rating data are available at `https://github.com/domino721/alzheimer-rag-evaluation`. The arXiv ancillary directory contains pseudonymised evaluator-level scores (2,000 records), a text-free table of expert averages and GPT scores (400 query–condition pairs), and executable scripts that reproduce the descriptive means, inter-rater reliability, rank associations, and query-cluster bootstrap. Stable query and rater identifiers support linkage without including evaluator names, query text, generated responses, or retrieved abstract text. These materials make the statistical reanalysis self-contained; they do not reproduce the original model calls.

The legacy pipeline is archived at `https://github.com/yeomsee/best_practices_bio_rag`, commit `25be531c6e8b5aff76b1a99a9011665e062299e2`. The corpus and DPR index are hosted at `https://huggingface.co/datasets/stv10121/pubmed_alzheimer_kb`. Exact model snapshots, execution dates, corpus deduplication and chunking parameters, and the final post-processing document count remain incompletely documented. No claim of exact end-to-end reproduction is made.

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